LECTURE 6

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In this lecture, we cover the contents of [BSY22, Section 8]. We discuss the application of the Nullstellensatz. We will use the Nullstellensatz theorem to explain orientations of Lubin-Tate theory and compute the strict picard spectrum of algebraic closed Lubin-Tate theories.

1. Pullback squares

Recall that by the first theorem in the last talk, for any none-zero $R \in \mathrm{CAlg}(\mathrm{Sp}T(n))$, there exists a map $R \to E(L)$ in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$. We can base change their modules in $\mathrm{Sp}_{T(n)}$ along this extension. The main theorem of this talk is

Theorem 1.1. *Suppose L is an algebraic closed field and $R \in \mathrm{CAlg}^{\wedge}_{E(L)}$. Then the base change $\mathrm{picMod}^{\wedge}_{E(L)} \to \mathrm{picMod}^{\wedge}_{R}$ admits a retract.*

We will heavily use the group of unit functor $\mathfrak{gl}_1 \colon \mathrm{CAlg} \to \mathrm{Sp}_{\geqslant 0}$. Recall that $\Omega\mathrm{pic}(R) = \mathfrak{gl}_1(R)$.

Moreover, we have $\mathfrak{gl}_1 E(-) : \mathrm{Perf}_k \to \mathrm{Sp}_{\geqslant 0}$ with $\mathfrak{gl}_1\pi_0 E(-)/m$ being exactly the group of units. We define $\mathfrak{sl}_1^E \in \mathrm{Sp}_{\geqslant 0}$ as the fiber of $\pi_0(-)/\mathfrak{m} : \mathfrak{gl}_1 E(-) \to \mathfrak{gl}_1(\pi_0 E(-)/\mathfrak{m})$

Proposition 1.2. *Suppose that we have a pullback in Perf_k.*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

such that $B \oplus C \to D$ is a surjection. Then after applying $\mathfrak{sl}_1^E$, we obtain a pullback square in $\mathrm{Sp}_{\geqslant 0}$:

$$\begin{array}{ccc} \mathfrak{sl}_1^E A & \longrightarrow & \mathfrak{sl}_1^E B \\ \downarrow & & \downarrow \\ \mathfrak{sl}_1^E C & \longrightarrow & \mathfrak{sl}_1^E D \end{array}$$

Proof. The construction of $\mathfrak{sl}_1^E$ includes applying the Lubin-Tate functor, the group of unit functor and the fiber to a given pullback square. All of these operators preserve pullbacks. To eliminate the "$\geqslant 0$" part, we need to show that $\pi_0(\mathfrak{sl}_1^E C) \oplus \pi_0(\mathfrak{sl}_1^E B) \to \pi_0(\mathfrak{sl}_1^E D)$ is a surjection. This follows from the surjectivity of $B \oplus C \to D$ and the complete filtration powered by $\mathfrak{m}$ on both sides. □

Corollary 1.3. *Suppose that we have a pullback square*

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

such that

(1) $B \oplus C \to D$ is a surjection.
(2) $B^\times \oplus C^\times / A^\times \to D^\times$ admits a retract.
(3) $A \to B$ and $A \to C$ admit retracts.

Then $\mathfrak{gl}_1(A) \to \mathfrak{gl}_1(D)$ admits a retract.

Proof. Note that (3) implies that we have a retract for

$$\mathfrak{gl}_1 E(A) \to \mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C)$$

Therefore, it suffices to find a retract for

$$\mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C) \to \mathfrak{gl}_1 E(D)$$

This follows from the above proposition, which implies that we have a pullback square

$$\begin{array}{ccc} \mathfrak{gl}_1 E(B) \coprod_{\mathfrak{gl}_1 E(A)} \mathfrak{gl}_1 E(C) & \longrightarrow & B^\times \coprod_{A^\times} C^\times \\ \downarrow & & \downarrow \\ \mathfrak{gl}_1 E(D) & \longrightarrow & D^\times \end{array}$$

and the fact that (2) implies right vertical arrow admits a splitting. □

2. Hahn series

Definition 2.1. Let Γ be a totally ordered commutative monoid and A be an integral domain. We define $A[\![t^\Gamma]\!]$ to be the set of functions $\Gamma \to A$ that have well-ordered support. It admits an obvious addition and a multiplication with formula

$$f \cdot g(x) = \sum_{y+z=x} f(y)g(z)$$

under which $A[\![t^\Gamma]\!]$ becomes a ring.

Lemma 2.2. *(1) $f \in A[\![t^\Gamma]\!]$ is invertible if and only if $\nu(f)$ is invertible and $\operatorname{arc}(f)$ is invertible, where the two maps are given by*

$$\begin{aligned} \nu \colon A[\![t^\Gamma]\!]^* &\to \Gamma & \qquad \operatorname{arc} \colon A[\![t^\Gamma]\!]^* &\to A^* \\ f &\mapsto \min_{f(\gamma)\neq 0} \gamma & f &\mapsto f(\nu(f)) \end{aligned}$$

(2) If A is a field, Γ is a group, then $A[\![t^\Gamma]\!]$ is a field.
(3) Let L be an algebraic closed field and Γ be a $\mathbb{Q}$-module, then $L[\![t^\Gamma]\!]$ is an algebraic closed field.

Corollary 2.3. *Let L be an algebraic closed field and F be a non-zero L-algebra. Suppose that Γ is a totally ordered $\mathbb{Q}$-module with $|\Gamma| > |F|$, then there exists an L-algebra map $F \to L[\![t^\Gamma]\!]$.*

Proof. By modulo the maximal ideal and taking the algebraic closure, we reduce to the case where F is algebraic closed. We have

$$|L[\![t^\Gamma]\!]| \geqslant |\Gamma| > |F|$$

Therefore, we conclude $\mathrm{tr}_L(|L[\![t^\Gamma]\!]|) > \mathrm{tr}_L(|F|)$ and there exists an inclusion $F \to L[\![t^\Gamma]\!]$. □

Lemma 2.4. *Let $A \in \mathrm{Perf}_k$ be a non-zero domain and Γ be a totally ordered monoid such that $p : \Gamma \to \Gamma$ is an isomorphism. Then the map*

$$\mathfrak{gl}_1 E(A) \to \mathfrak{gl}_1 E(A[\![t^\Gamma]\!])$$

admits a retract.

Proof. This follows from applying Corollary 1.3 to the pullback square

$$\begin{array}{ccc} A & \longrightarrow & A[\![t^{\Gamma_{\leqslant 0}}]\!] \\ \downarrow & & \downarrow \\ A[\![t^{\Gamma_{\geqslant 0}}]\!] & \longrightarrow & A[\![t^\Gamma]\!] \end{array}$$

and the description of the invertible elements in $A[\![t^\Gamma]\!]$. □

Now we can prove our main theorem.

Theorem 2.5. *Suppose L is an algebraic closed field and $R \in \mathrm{CAlg}^\wedge_{E(L)}$. Then the base change $\mathrm{picMod}^\wedge_{E(L)} \to \mathrm{picMod}^\wedge_R$ admits a retract.*

Proof. By the existence of extension $F \to L[\![t^\Gamma]\!]$ and the Nullstellensatz theorem, we reduce to the case where $R = L[\![t^\Gamma]\!]$. We consider the following square

$$\begin{array}{ccc} \Sigma\mathfrak{gl}_1 E(L) & \longrightarrow & \Sigma\mathfrak{gl}_1 E(L[\![t^\Gamma]\!]) \\ \downarrow & & \downarrow \\ \mathrm{pic}E(L) & \longrightarrow & \mathrm{pic}E(L[\![t^\Gamma]\!]) \end{array}$$

This square is a pushout square since they admit isomorphic cofibers. By Lemma 2.4, the above arrow has a retract, which yields a retract for the bottom arrow. □

3. Orientations and Thom spectra

Definition 3.1. Let $\mathcal{C}$ be a symmetric monoidal category and $X \to \mathrm{pic}(\mathcal{C})$ be a map in $\mathrm{Sp}_{\geqslant 0}$. We define the *Thom spectrum* of f to be $Mf := \mathrm{colim}_X f \in \mathrm{CAlg}(\mathcal{C})$, where the colimit is taken in $\mathcal{C}$.

According to [And+14], Mf satisfies a specific universal property. That is, the space of orientations from Mf to R can be identified with the lifting of f to trivial R-modules.

Proposition 3.2. *Let $\mathcal{C}$ be a presentable symmetric monoidal category. Then the followings are equivalent.*

(1) $f : X \to \mathrm{pic}\mathcal{C}$ is null.

(2) There is a map $Mf \to \mathbb{1}_{\mathcal{C}}$ *in* $\mathrm{CAlg}(\mathcal{C})$.
(3) $\mathrm{pic}(\mathcal{C}) \to \mathrm{pic}(\mathrm{Mod}_{Mf}(\mathcal{C}))$ *admits a retract.*

Corollary 3.3. *Let* L *be an algebraic closed field and* $f\colon X \to \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})$ *be a map in* $\mathrm{Sp}_{\geqslant 0}$. *The the followings are equivalent:*

(1) f *is null.*
(2) There is a map $Mf \to E(k)$.
(3) Mf *is nonzero.*
(4) $Mf \simeq E(L)[X]$.

4. Strict Picard group of Lubin-Tate theory

The above discussion can be used to compute the strict Picard group of Lubin-Tate theory.

Definition 4.1. Let $\mathcal{C}$ be a presentable symmetric monoidal category. The strict Picard spectrum of $\mathcal{C}$ is defined as

$$\mathbb{G}\mathrm{pic}(\mathcal{C}) = \underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathcal{C})) \simeq \tau_{\geqslant 0}\mathrm{map}_{\mathrm{Sp}_{\geqslant 0}}(\mathbb{Z}, \mathrm{pic}(\mathcal{C}))$$

Proposition 4.2. *Let* L *be an algebraically closed field and let* H *be a* p*-torsion abelian group. There is a natural equivalence of connective spectra*

$$\underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geq 0}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})) \simeq \Sigma^{n+1} H^*$$

where $H^* = \mathrm{Hom}(H, \mathbb{Q}/\mathbb{Z})$ *stands for Pontryagin dual of* H.

Proof. (Sketch of proof) We first show that $\pi_m \mathrm{Map}_{\mathrm{Sp}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})) \cong 0$ for $m \neq n+1$. It suffices to show that a class

$$f \in \pi_0(\mathrm{Map}_{\mathrm{Sp}}(\Sigma^m H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)}))$$

is null with $0 \leqslant m \leqslant n$. By Corollary 3.3, it suffices to show that $Mf \neq 0$, which is by argument in [HL13]. Now we need to identify the functor $\pi_n \underline{\mathrm{Hom}}_{\mathrm{Sp}_{\geq 0}}(H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)}))$ with the Pontryagin dual $(-)^*$. This can be check on the set level, which is given by

$$\begin{aligned}[\Sigma^{n+1} H, \mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(L)})] &\cong [\Sigma^n H, \mathfrak{gl}_1(E(L))] \cong [E(L)[B^n H], E(L)]_{\mathrm{CAlg}^{\wedge}_{E(L)}} \\ &\cong [E(L)^{H^*}, E(L)]_{\mathrm{CAlg}^{\wedge}_{E(L)}} \cong H^*\end{aligned}$$

□

Theorem 4.3. *Let* L *be an algebraically closed field of characteristic* p *and assume that the height* $n \geqslant 1$ *Then we have an equivalence of connective* $\mathbb{Z}$*-modules*

$$\mathbb{G}\mathrm{pic}(\mathrm{Mod}^{\wedge}_{E(k)}) \simeq \Sigma^{n+2}\mathbb{Z}_p \oplus \Sigma L^{\times}$$

References

[And+14] Matthew Ando et al. "An ∞-categorical approach to R-line bundles, R-module Thom spectra, and twisted R-homology". In: *Journal of Topology* 7.3 (2014), pp. 869–893 (cit. on p. 3).

[BSY22] Robert Burklund, Tomer M Schlank, and Allen Yuan. "The chromatic nullstellensatz". In: *arXiv preprint arXiv:2207.09929* (2022) (cit. on p. 1).

[HL13] Michael Hopkins and Jacob Lurie. "Ambidexterity in $K(n)$-local stable homotopy theory". In: *preprint* (2013), p. 25 (cit. on p. 4).